



Ethernet for Shared AI Fabrics

Full Performance with Multi-Tenant Scale

Will Eatherton

SVP, Data Center, Internet & Cloud
Infrastructure Engineering

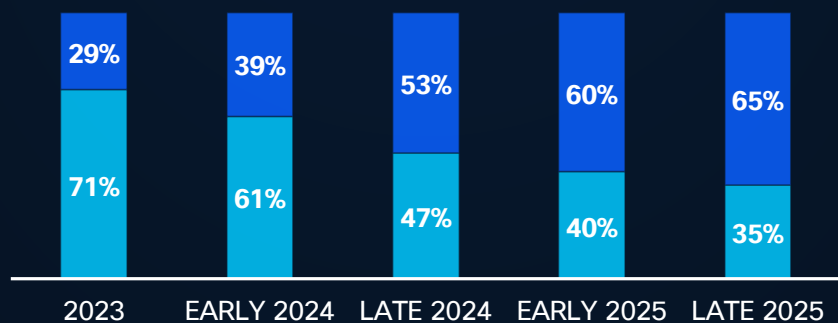


July 2026

The industry shift toward Ethernet for AI networking

AI Networks Market Share %

■ Infiniband ■ Ethernet



Who's Already On Ethernet

Meta: Trained Llama 3 on 24,576 GPUs using RoCE Ethernet. Peer-reviewed at ACM SIGCOMM 2024: equivalent performance

xAI Colossus: 100,000 GPU cluster, the world's largest single AI training job, built entirely on Ethernet

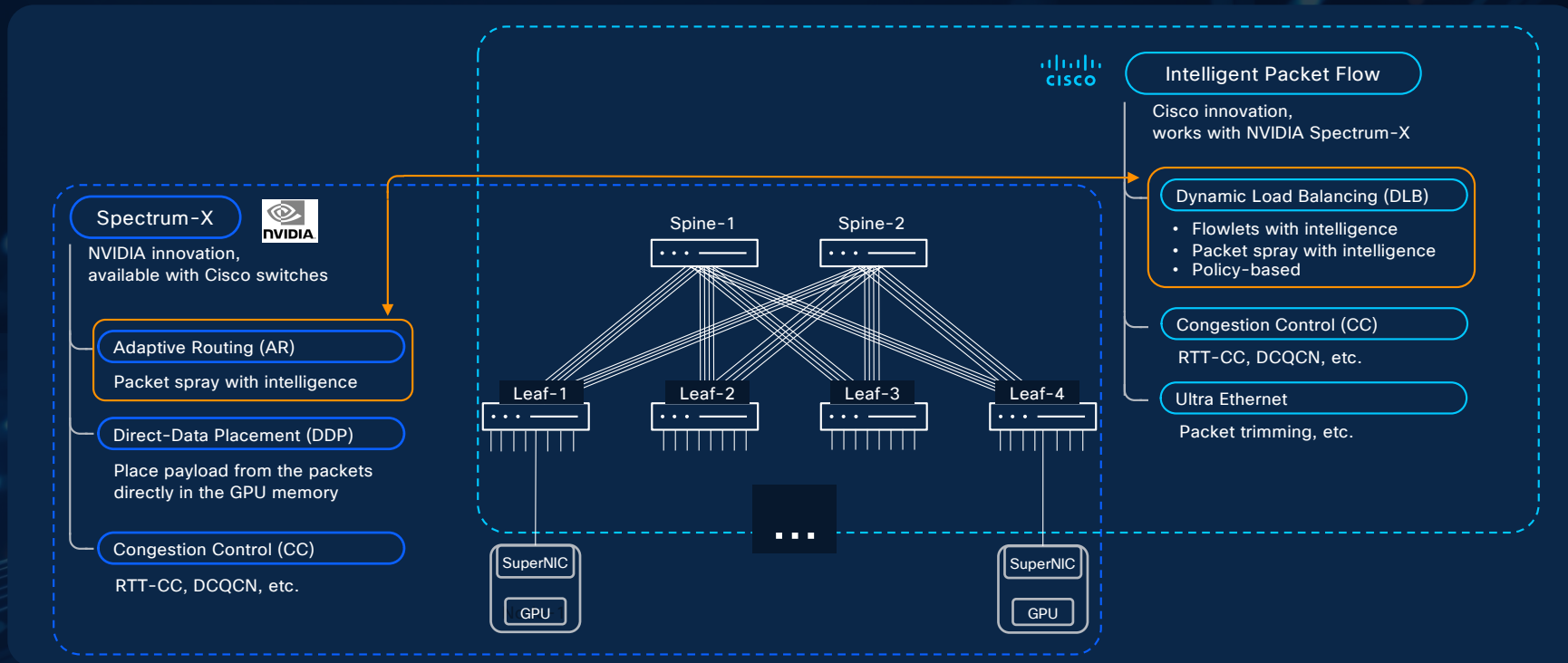
Google, Amazon, Microsoft, Oracle: All identified by Dell'Oro Group as driving Ethernet adoption in AI back-end networks through 2025

NVIDIA: Launched Spectrum-X, an AI-optimized Ethernet platform; Investing in Ethernet silicon photonics shipping 2026

Sources: 650 Group Data Center Networking - AI - By Technology Report 1Q26 | Meta Engineering: "Building Meta's GenAI Infrastructure" (Mar 2024), "RoCE for Distributed AI Training at Meta Scale", ACM SIGCOMM 2024 | HPCwire: xAI Colossus (Sep 2025) | NVIDIA GTC 2025 Keynote: Spectrum-X announcement (Mar 2025)

Intelligent load balancing and congestion control

Making the Ethernet AI fabric lossless and low latency

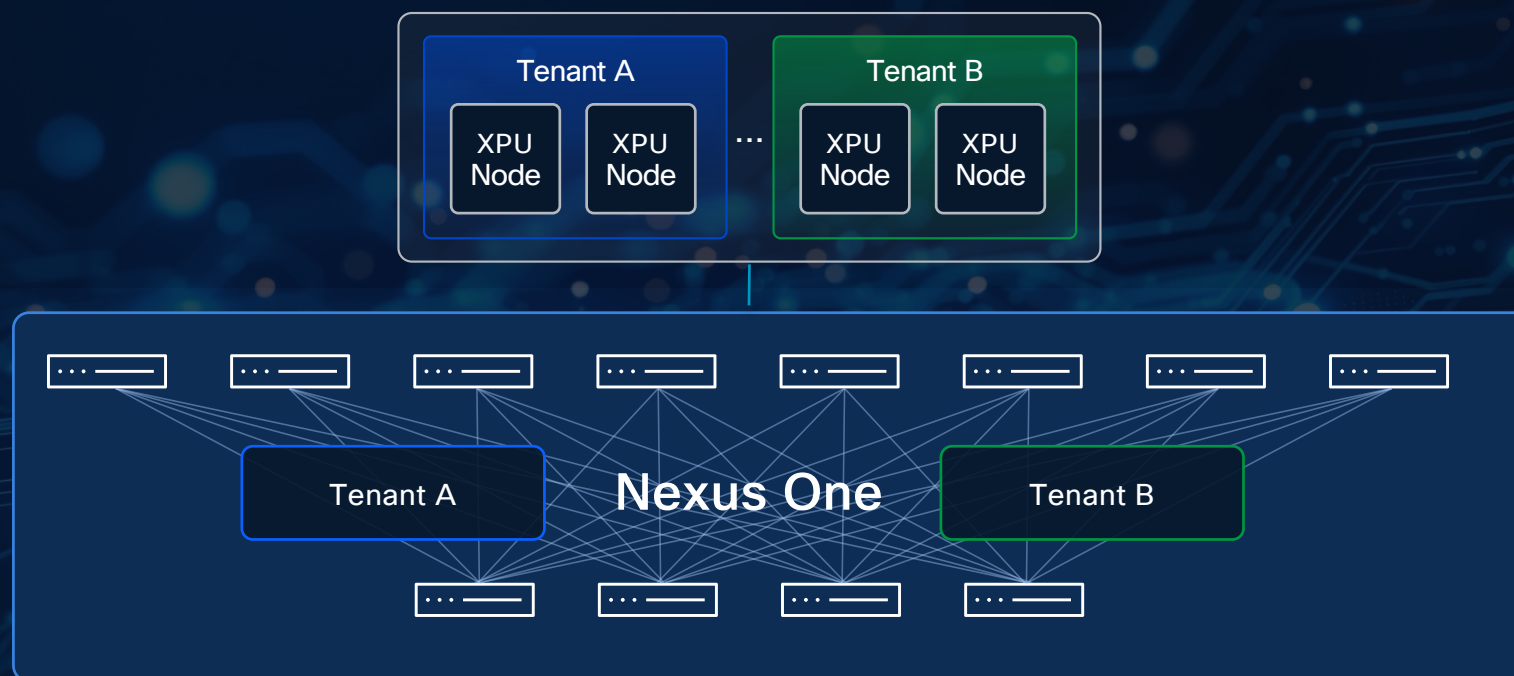


Ethernet gain

Measured, signaled, adaptive fabric behavior

Multi-tenancy in the backend AI network

Turning a dedicated AI fabric into a shared one



AI Orchestration Integration with Nexus One



Solution

Automated Tenant
Onboarding

Multi-tenant isolation
at scale

Better monetization

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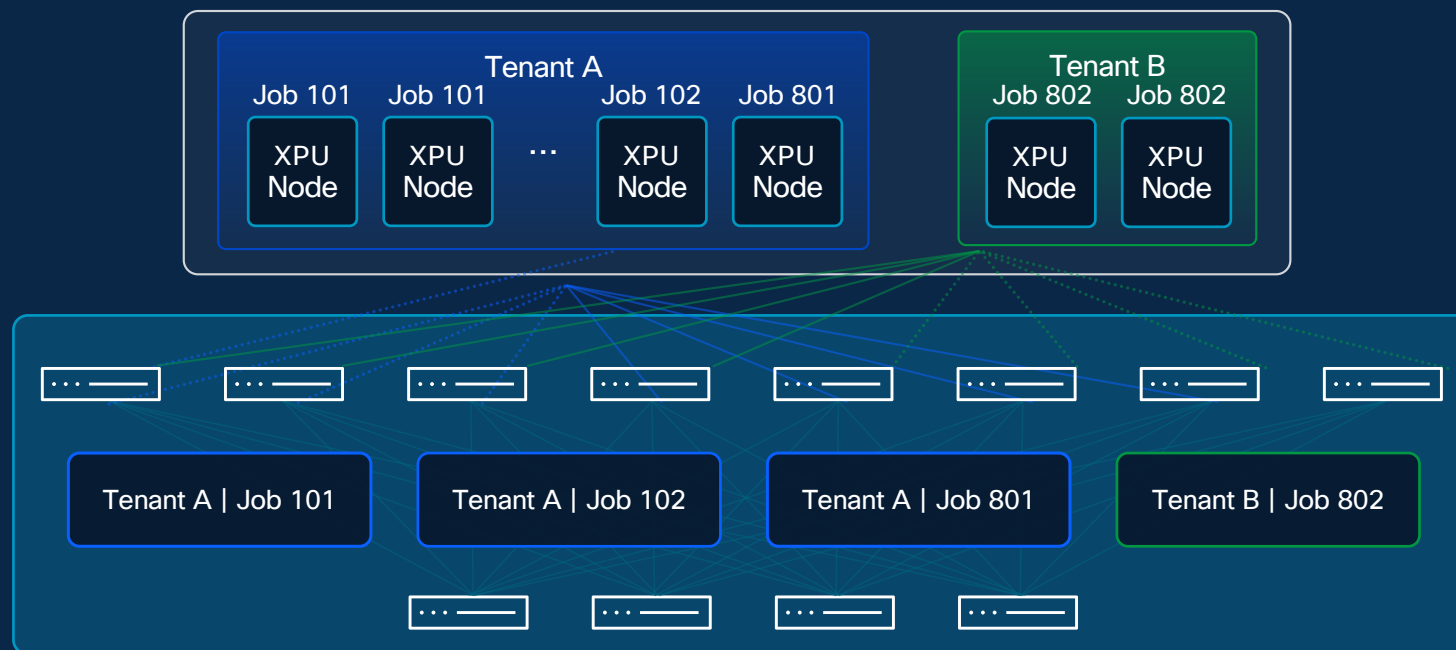
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From tenant-level to job-level segmentation

NEW GA Q4 CY26

Giving workloads protected lanes



Ethernet gain

Shared infrastructure economics with job-level guardrails

Scale-out provides high NCCL efficiency leading to optimal JCT

NCCL BUSBW | Full common sweep | 128 H100 GPUs

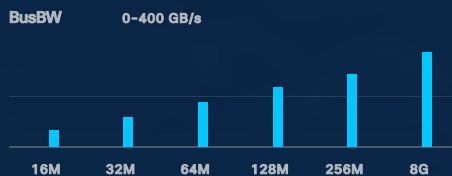
NCCL Bus Bandwidth scales linearly with message size

■ Cisco Ethernet (G200)

All reduce

BusBW by message size

Cisco G200 4x2 RO



All to all

BusBW by message size

Cisco G200 4x2 RO



MLPERF training v6.0 | LLAMA 3.1 8B

Cisco submission

21.97 min

TO 3.3 Log-perplexity target

16

NODES

128

H100 GPUs

400 Gb/s

PER GPU RoCEv2

VALIDATED FABRIC

RAIL-OPTIMIZED RoCEv2 | CISCO SILICON ONE G200

8 GPUs/NODE | ONE 400 Gb/s NIC PER GPU

Why it matters

Accelerates your AI training with an open, high-performance Ethernet fabric built to scale

Looking forward

SRv6 makes AI fabrics deterministic

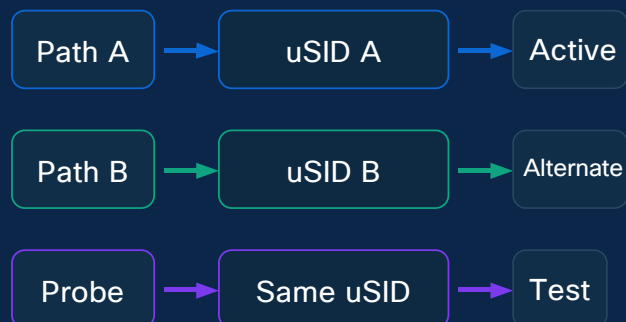
Why SRv6 was chosen

Not just packet spraying

- 1 Deterministic placement**
Each packet or spray round can be pinned to a specific uSID program
- 2 Stateless static fabric**
Leaf and spine devices execute local SRv6 instructions; no per-flow state or path install
- 3 No convergence dependency**
When a path degrades, the source changes the uSID program while intermediate switches stay unchanged.
- 4 Ground-truth visibility**
SRv6 probes follow the same physical path as data, removing ECMP ambiguity for operations

SRv6 fabric model

Path catalog created at bring-up; packets carry the selected uSID program



Fabric principle

Change paths by selecting another uSID program

MRC reference architecture

- Extends RoCE with multipath packet spraying
- Maps path IDs to SRv6 uSID programs
- Validates static SRv6 source routing in 100k+ GPU-class AI fabrics

Cisco contribution to SRv6

- Inventor / primary architect of SRv6
- Co-authored core SRv6 standards: RFC 8402, RFC 8754, RFC 8986
- Collaboration with lead operators to bring technology to maturity

The accelerating trajectory of Ethernet for AI

Open standards, hyperscaler investment, and MRC deployments are securing Ethernet as the foundation for AI infrastructure

